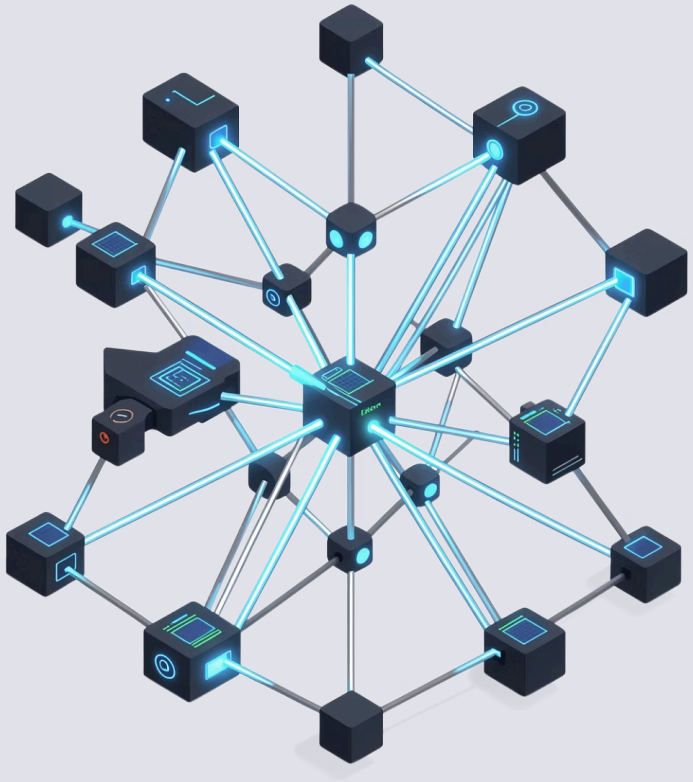
Graph Neural Networks (GNNs)

Exploration of GNNs and Their Applications in Modeling Relational Data

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What is a Graph? — The Building Block

Core Concepts

Node (Vertex)

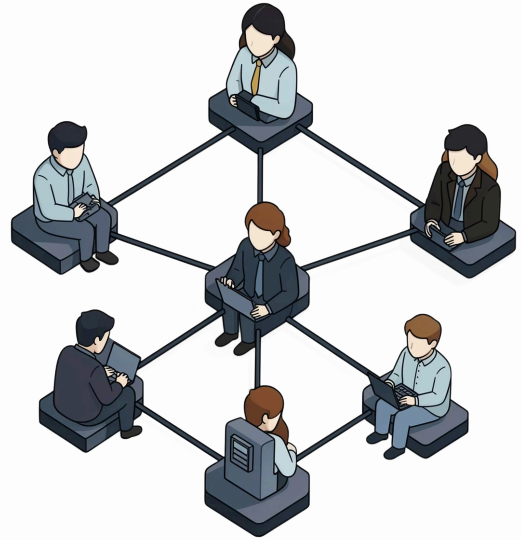
Represents an entity — a person, atom, or webpage

Edge (Link)

Represents a relationship — friendship, bond, or hyperlink

Example

Facebook — users are nodes, friendships are edges



Think of a graph as a map of "**who is connected to whom.**"

Why Can't Normal Neural Networks Handle Graphs?

Traditional neural networks are designed for **fixed structures** — but graphs are irregular and flexible.

🖼️ CNNs

Work on **grids** (images)

Fixed pixel neighbors — always a 3×3 or 5×5 grid around each pixel

📝 RNNs

Work on **sequences** (text)

Fixed order — one word follows the next in a line

🔗 Graphs

Are **irregular** — each node can have any number of neighbors

No fixed grid or sequence structure exists

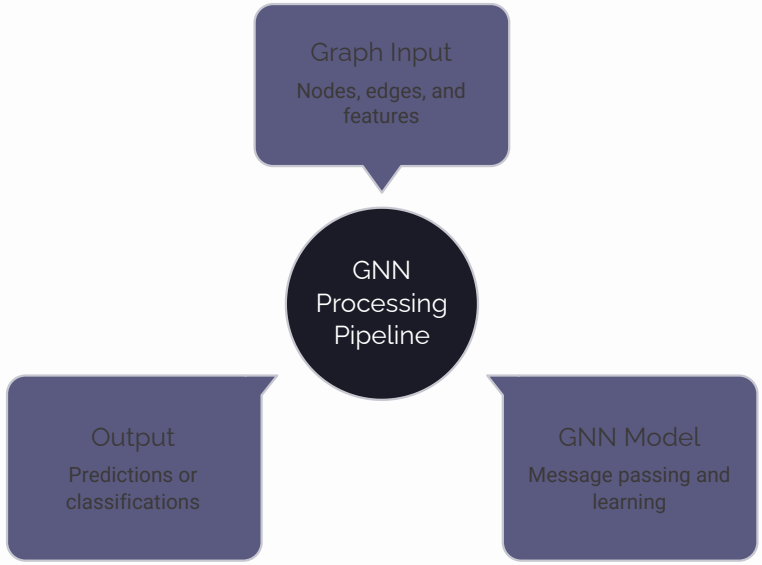
ⓘ We need a **special network** that understands relationships, not just individual data points. That's where GNNs come in!

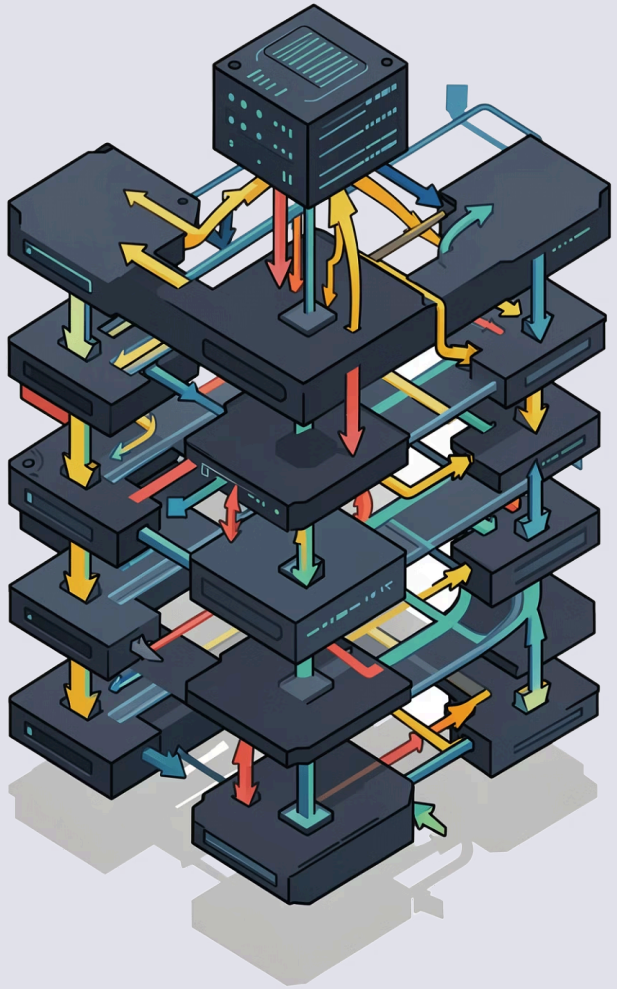
What is a Graph Neural Network (GNN)?

Definition

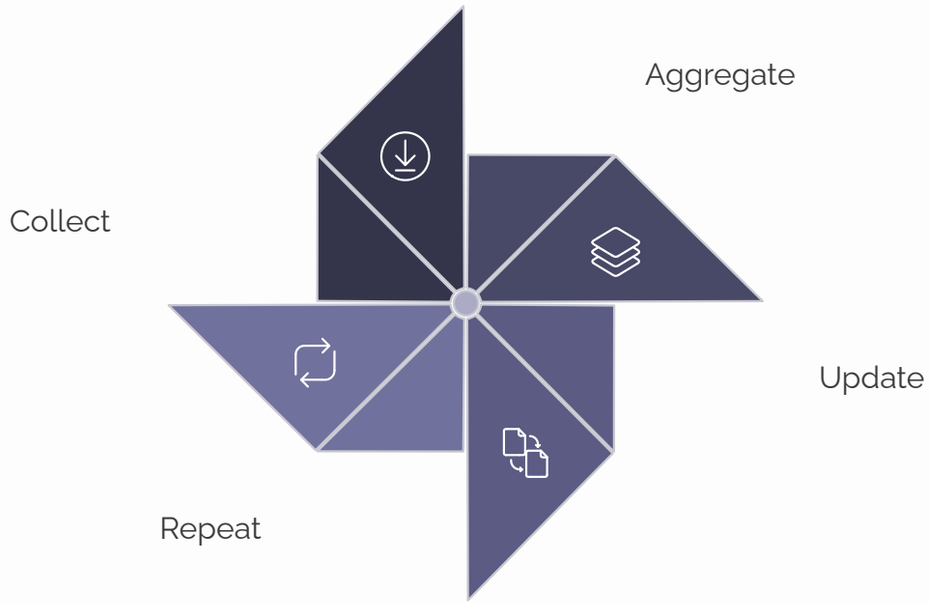
A type of deep learning model designed to work directly on **graph-structured data**. GNNs learn from both **node features** (what a node is) and **edges** (who it connects to).

They capture hidden patterns in relationships that traditional ML simply cannot see.





How Does a GNN Learn? — Message Passing



The Core Learning Mechanism

1 Collect

Each node gathers information from all of its direct neighbors

2 Aggregate

Combines neighbor info with its own existing features

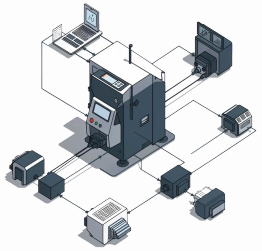
3 Update

Updates its own representation — called an **embedding**

Repeated over multiple layers → a node "sees" farther and farther across the graph

Types of GNNs — A Quick Overview

Three popular GNN architectures, each with a different approach to learning from neighbors:



GCN

Graph Convolutional Network

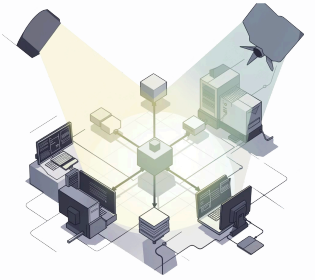
Averages all neighbor features — simple, fast, and great for beginners



GraphSAGE

Sample and Aggregate

Samples a subset of neighbors — scales efficiently to very large graphs

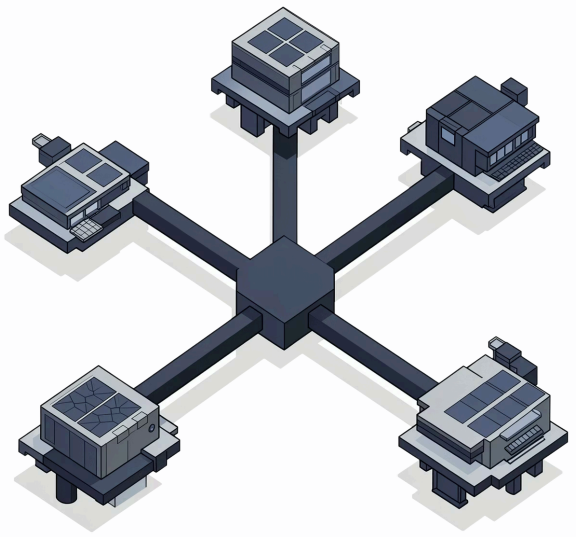


GAT

Graph Attention Network

Pays more attention to the most important neighbors — smarter aggregation

GNN Tasks — What Can GNNs Predict?



Three Core Prediction Tasks

Node Classification

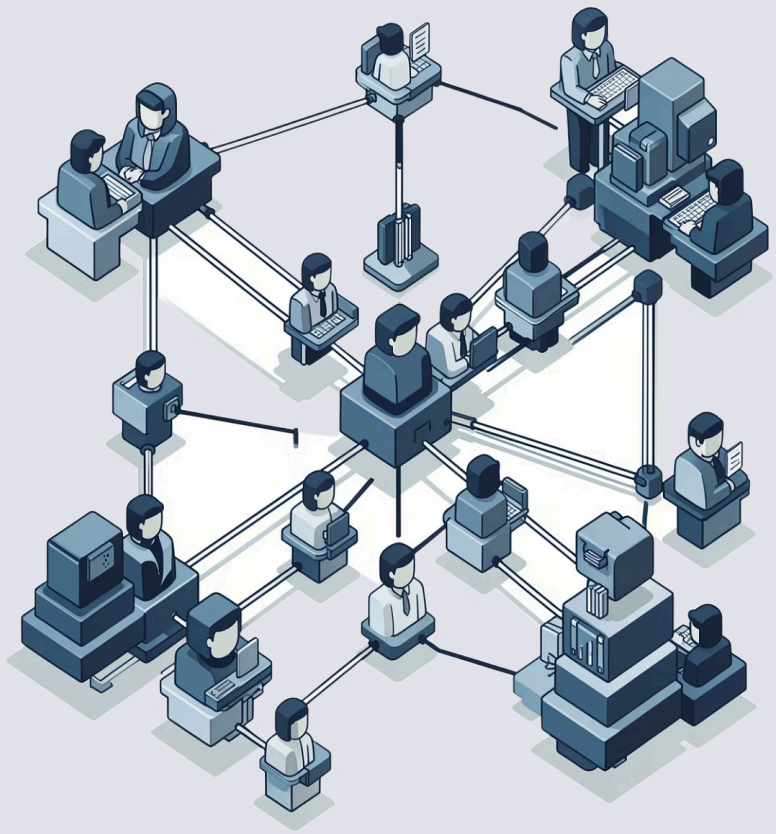
Is this user a spammer? GNN assigns a label to each individual node in the graph

Link Prediction

Will these two users become friends? GNN predicts whether a new edge should exist

Graph Classification

Is this molecule toxic? GNN labels the entire graph as a whole



Application 1 — Social Networks

Community Detection

Find groups of closely connected users — identify friend circles and interest clusters

Fake Account Detection

Spot bots by unusual connection patterns — flagged by abnormal edge behavior

Influence Prediction

Who will spread information the fastest? Model information flow through the network

Application 2 — Recommendation Systems

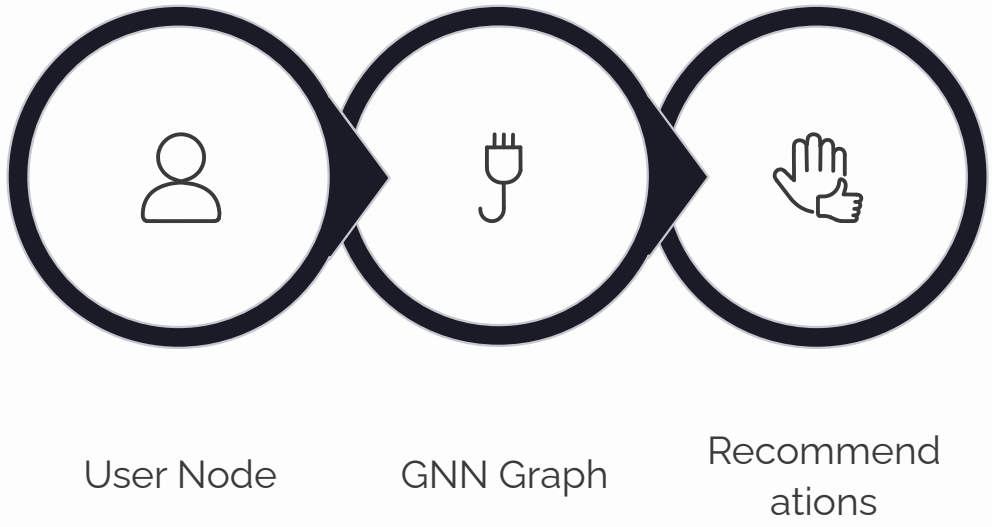
How GNN-Powered Recommendations Work

In a recommendation graph, **nodes** represent users and products. **Edges** represent purchases, ratings, and clicks.

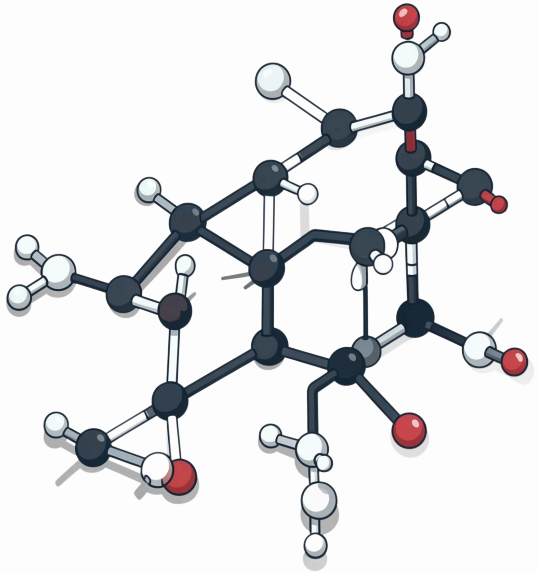
The GNN learns user preferences by examining what **similar users** liked — surfacing the most relevant content for each person.

PINTEREST YOUTUBE AMAZON

All three platforms use GNN-powered graphs to power their recommendation engines.



Application 3 — Biological Networks



GNNs in Biology & Drug Discovery

Protein Interaction Networks

Nodes = proteins, Edges = interactions. GNNs map how proteins work together in living cells

Drug Discovery

GNNs predict which molecules can treat diseases by analyzing molecular graphs of atoms and bonds

AlphaFold-Style Research

Predict protein structure from a graph of atoms — a breakthrough in computational biology

Real-World GNN Use Cases at a Glance

🚦 Traffic Prediction

Google Maps uses GNNs to predict travel time by modeling roads as a graph of intersections and routes

💳 Fraud Detection

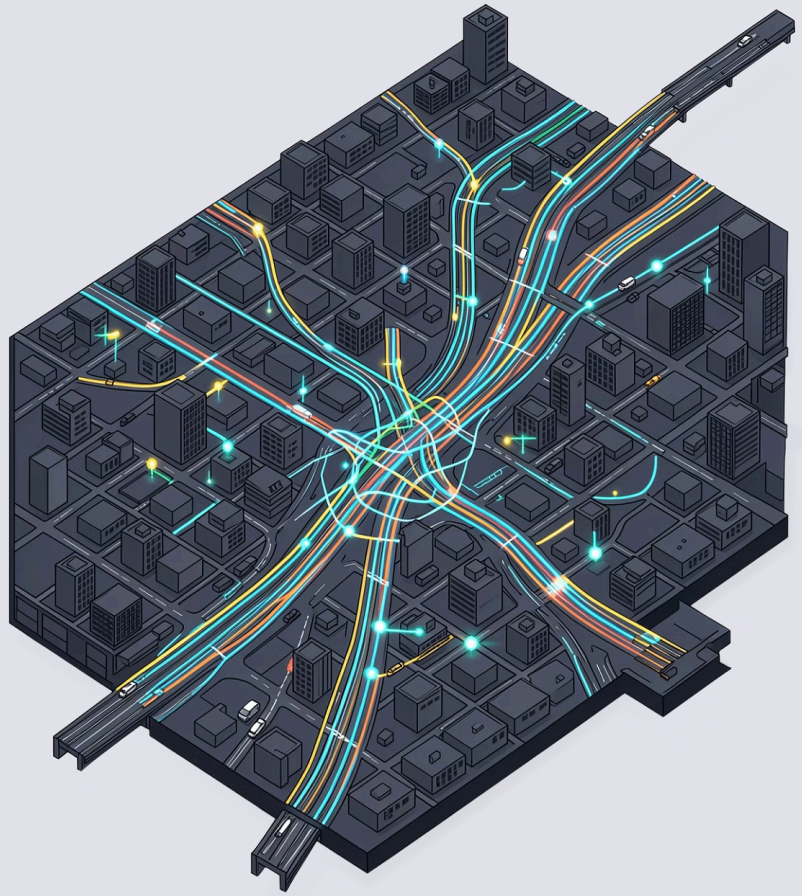
Banks detect suspicious transaction patterns by analyzing the graph of accounts, transactions, and merchants

📰 Fake News Detection

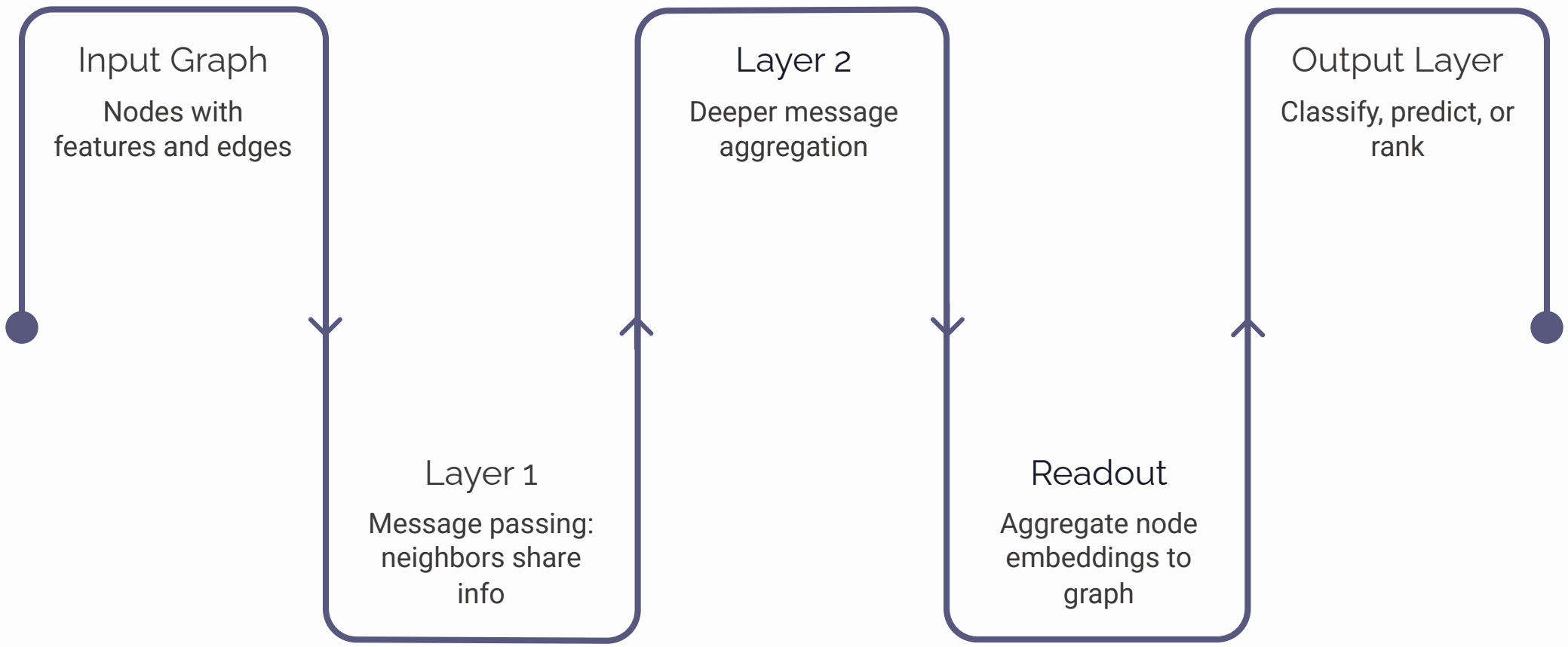
Identify misinformation as it spreads across social graphs — catching it before it goes viral

🧬 Antibacterial Discovery

Find new antibiotics by analyzing molecular graphs — GNNs have already identified novel drug candidates



GNN Architecture — End-to-End Flow



1

Input

Graph with node features and edge connections

2

GNN Layers

Multiple rounds of message passing

3

Readout

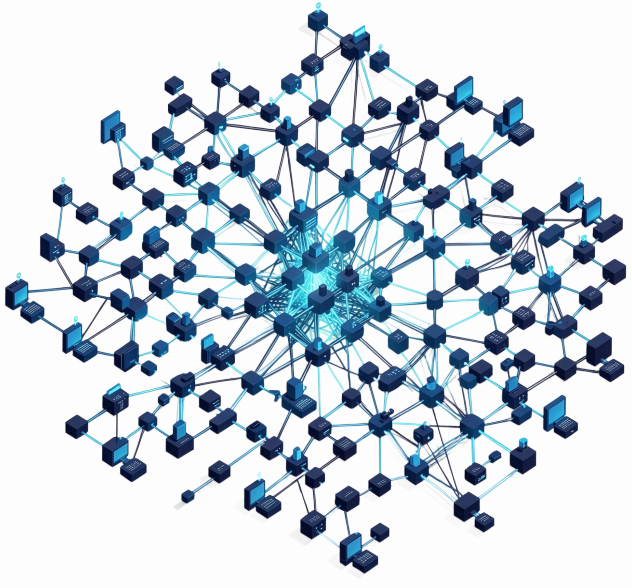
Aggregate all node embeddings into a graph-level vector

4

Output

Classify, predict, or rank

Advantages of GNNs



Why GNNs Are Powerful

✅ Handles Irregular Data

Naturally handles relational data with no fixed structure needed — no grid, no sequence required

✅ Dual Learning

Learns from both **node content** (what something is) and **graph structure** (how it connects)

✅ Scales Massively

Designed to scale to **millions of nodes** — like the scale of real social networks

✅ Generalizes Well

Works across different graph sizes and shapes — one trained model, many applications

Limitations of GNNs

Like all models, GNNs have real challenges that researchers are actively working to solve:

⚠️ Over-Smoothing

Too many GNN layers make all node embeddings look the same — nodes lose their unique identity as information blends together across too many hops

⚠️ Scalability

Very large graphs are computationally expensive — storing and processing billions of edges requires specialized hardware and algorithms

⚠️ Dynamic Graphs

GNNs struggle when edges or nodes change over time — most architectures assume the graph is **static**

⚠️ Interpretability

Hard to explain *why* a GNN made a specific decision — a significant challenge for high-stakes applications like medicine and law

GNNs vs Traditional Machine Learning

How do GNNs compare to conventional ML approaches for relational data tasks?

Feature	Traditional ML	Graph Neural Networks
Data Type	Tabular / Grid	Graph (Nodes + Edges)
Captures Relationships	❌ No	✅ Yes
Works on Social Networks	❌ Limited	✅ Naturally
Handles Irregular Structure	❌ No	✅ Yes
Example Use	Image Classification	Friend Recommendation



Key Takeaways & Next Steps

- [ ] GNNs are the go-to tool for relational, graph-structured data

  Wherever there are entities and relationships, GNNs can find patterns

- [ ] Core idea: message passing — learn from your neighbors

  Collect → Aggregate → Update, repeated across multiple layers

- [ ] Applications span social networks, recommendations, biology, and more

  From Google Maps to drug discovery — GNNs are everywhere

- [ ] Start learning: PyTorch Geometric or DGL — both free & beginner-friendly

  Hands-on practice is the fastest path to mastery

Dr. Anilkumar V Brahmane | Computer Engineering Department, Sanjivani College of Engineering

Sources

IBM Think

What is a Graph Neural Network | IBM

NVIDIA Research

Introduction to Graph Neural Networks | NVIDIA Structured Data and Graph Models

Machine Learning Knowledge

Graph Neural Networks (GNN) Explained for Beginners — MLK Machine Learning Knowledge